

University of the Witwatersrand
School of Computer Science and Applied Mathematics
COMS4054A & COMS7062A & COMS7066A:
Natural Language Processing
Practice Quiz

The following are example questions used to demonstrate the kinds of questions which will come up in the quiz. While an accurate demonstration of the standard, these questions are not comprehensive and cannot be used in isolation as preparation for the quiz. Additionally, I have separated questions into sections to make it easy to look for the answers and identify your weaker sections. In the quiz on moodle the order will be random. Lastly, the given mark indications are a guide but it is possible that similar questions may have different mark allocations in the quiz based on their difficulty.

1 Language Basics

1. Syntax is the rule system of how to combine: (1 Mark)
 - (a) Letters to form morphemes
 - (b) Morphemes to form words
 - (c) Words to form sentences
 - (d) Sentences to form paragraphs
2. The difference between a root and stem word is: (1 Mark)
 - (a) The stem is the root word with derivational affixes
 - (b) They are completely different words in meaning
 - (c) The stem is the root word with inflectable affixes
 - (d) Root word is a synonym but different word to the stem
3. True or False, when we reference “semantics” we are referring to the meaning of words and how they combined to form meaningful sentences. (1 Mark)

2 Basic Processing and Vector Representations

1. Which of the following are true? (4 Marks)
 - (a) Word2Vec performs negative sampling as a means to speed up training.
 - (b) Word2Vec embeddings are more useful for downstream tasks than 1-hot representations on words which aren't in the vocabulary
 - (c) Intrinsic evaluation of word embeddings evaluates how well the word vectors perform on a downstream task
 - (d) Simple tokenization schemes are particularly good for languages with rich morphology
2. Complete the sentence: "The problem with using the cosine distance to measure word similarity of vector representations is..." (1 mark)
 - (a) the cosine distance does not account for the magnitude of the vectors (only the angle between them)

- (b) the cosine distance is longer if words have higher frequencies
- (c) the cosine distance is longer if words have higher values
- (d) the cosine distance is 0 when vectors are orthogonal

3 Text Classification

1. What function is typically used to get the probability distribution over classes, after the linear layer? Please write the name of the function only without spaces at the beginning and end of the name. (1 mark)
2. True or False, the Bag-of-Words (count based features) is an example of a neural method of feature extraction. (1 Mark)
3. Laplace smoothing is used in case: (1 Mark)
 - (a) Too many stop words appear in the training data.
 - (b) An unseen word appears during inference.
 - (c) The class labels are very imbalance.
 - (d) All of the above.
4. True or False, a support vector machine (SVM) is a discriminative model similar to logistic regression (can be used for classification). (1 Mark)
5. True or False, CNNs for NLP represent time spatially while RNNs do not. (1 Mark. * this is probably on the more difficult side and might require some thinking)

4 Language Modelling

1. True or False, an embedding layer is functionally equivalent to a linear layer (1 Mark)
2. True or False, when using an N-gram language model Backoff aims to fix the problem that, as N becomes larger, individual inputs (the N-grams) become more rare (1 Mark)
3. Please select the answers that are true statements about text generation strategies (5 Marks)
 - (a) Temperature is a hyper-parameter added to the softmax of a neural language model
 - (b) By changing the temperature we aim to control the coherence-diversity trade-off in neural language models
 - (c) Higher temperatures result in more coherent text
 - (d) Lower temperatures result in more coherent text
 - (e) True or False, Nucleus sampling (Top-p sampling) is an adaptive method of sampling which adjust how many words could be chosen from based on how many words probable words there are

5 Seq2Seq, Attention and Transformers

1. Which of the statements is true about the structure of Seq2Seq Models:
 - (a) The encoder processes the previous history
 - (b) The decoder processes the source text
 - (c) The encoder provides word embeddings for the decoder
 - (d) The decoder processes the previous history
2. Which one of the following does the final encoder state of an RNN represent?
 - (a) The semantic embedding of the last word it is shown
 - (b) The semantic embedding of the entire sentence with a bias towards the last word it is shown
 - (c) The semantic embedding of the entire sentence with no positional bias
 - (d) The semantic embedding of the entire sentence with a bias towards the first word it is shown
3. True or False: Seq2Seq models predict next words in a sentence.
4. True or False: The use of Softmax in the attention calculation makes the attention sparse.
5. Which of the following is **not** a common score function used within attention (not necessarily only self attention in Transformers):
 - (a) Bilinear Attention
 - (b) Multi-Layer Perceptron
 - (c) Softmax Attention
 - (d) Dot-Product Attention